

Training Long-Context Vision-Language Models Effectively with Generalization Beyond 128K Context

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ABSTRACT

Long-context modeling is becoming a core capability of modern large vision-language models (LVLMs), enabling sustained context management across long-document understanding, video analysis, and multi-turn tool use in agentic workflows. Yet practical training recipes remain insufficiently explored, particularly for designing and balancing long-context data mixtures. In this work, we present a systematic study of long-context continued pre-training for LVLMs, extending a 7B model from 32K to 128K context with extensive ablations on long-document data. We first show that long-document VQA is substantially more effective than OCR transcription. Building on this observation, our ablations further yield three key findings: i) for sequence-length distribution, balanced data outperforms target-length-focused data (e.g., 128K), suggesting that long-context ability requires generalizable key-information retrieval across various lengths and positions; ii) retrieval remains the primary bottleneck, favoring retrieval-heavy mixtures with modest reasoning data for task diversity; and iii) pure long-document VQA largely preserves short-context capabilities, suggesting that instruction-formatted long data reduces the need for short-data mixing. Based on these findings, we introduce MMProLong, obtained by long-context continued pre-training from Qwen2.5-VL-7B with only a 5B-token budget. MMProLong improves long-document VQA scores by 7.1% and maintains strong performance at 256K and 512K contexts beyond its 128K training window, without additional training. It further generalizes to webpage-based multimodal needle retrieval, long-context vision-text compression, and long-video understanding without task-specific supervision. Overall, our study establishes a practical LongPT recipe and an empirical foundation for advancing long-context vision-language models.

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Abstract

Long-context modeling is becoming a core capability of modern large vision-language models (LVLMs), enabling sustained context management across long-document understanding, video analysis, and multi-turn tool use in agentic workflows. Yet practical training recipes remain insufficiently explored, particularly for designing and balancing long-context data mixtures. In this work, we present a systematic study of *long-context continued pre-training* for LVLMs, extending a 7B model from 32K to 128K context with extensive ablations on long-document data. We first show that long-document VQA is substantially more effective than OCR transcription. Building on this observation, our ablations further yield three key findings: i) for sequence-length distribution, balanced data outperforms target-length-focused data (e.g., 128K), suggesting that long-context ability requires generalizable key-information retrieval across various lengths and positions; ii) retrieval remains the primary bottleneck, favoring retrieval-heavy mixtures with modest reasoning data for task diversity; iii) pure long-document VQA largely preserves short-context capabilities, suggesting that instruction-formatted long data reduces the need for short-data mixing. Based on these findings, we introduce **MMPROLONG**, obtained by long-context continued pre-training from Qwen2.5-VL-7B with only a 5B-token budget. **MMPROLONG** improves long-document VQA scores by 7.1% and maintains strong performance at 256K and 512K contexts beyond its 128K training window, without additional training. It further generalizes to webpage-based multimodal needle retrieval, long-context vision-text compression, and long-video understanding without task-specific supervision. Overall, our study establishes a practical LongPT recipe and an empirical foundation for advancing long-context vision-language models.

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1 Introduction

The ability to process long context has unlocked a wide range of new capabilities for both large language models [LLMs; 1, 2] and large vision-language models [LVLMs; 3, 4]. For LVLMs in particular, long-context modeling enables multi-hop reasoning over document collections [5, 6], capturing spatiotemporal dependencies from hour-long videos [7, 8], and maintaining context consistency in long-horizon agent tasks [9–11].

To support such capabilities, LVLMs’ context windows have been rapidly scaled to 128K tokens and beyond, driven by both proprietary models (e.g., Gemini 3.1 [12] and GPT-5.4 [13]) and open-weight alternatives such

as Qwen3-VL [3] and GLM-4.5V [14]. However, recent technical reports [3, 14] provide only limited details on the use of long-document data, leaving practical recipes for developing long-context vision-language models (LVLMs) insufficiently explored. It remains unclear which types of long-context data to synthesize, how to mix different long-context tasks and incorporate short-context data, and how training choices such as length distributions affect the resulting model.

To bridge this gap, we present a systematic study of the *long-context continued pre-training* (LongPT) in LVLMs. Building on Qwen2.5-VL-7B [15], we extend its context window from 32K to 128K and study how to construct and combine multimodal long-context training data. We use long documents as data sources because they provide realistic multimodal contexts with complex visual layouts and dense textual content. From these documents, we construct five training tasks grouped into two task categories: long-document VQA and OCR transcription. Comparing these tasks, we find that long-document VQA is substantially more effective than OCR transcription, suggesting that instruction-formatted supervision and task diversity ranging from information extraction to complex numerical reasoning are important for LongPT.

Having established long-document VQA as the primary data source, we then study practical training designs for LongPT in LVLMs, covering sequence-length distribution, long-context task mixtures, and the role of short-context data. In these ablations, we observe three main findings: i) for sequence-length distribution, we find balanced data outperforms target-length-focused data near 128K, suggesting that LongPT should teach generalizable key-information retrieval across various lengths and positions rather than specialize to a single target length; ii) key-information retrieval remains the primary bottleneck in long-context pre-training, favoring retrieval-heavy mixtures with modest reasoning data to maintain task diversity; and iii) unlike LLM long-context pre-training practice [16], pure long-document VQA largely preserves short-context capabilities, suggesting that instruction-formatted long data reduces the need for short-context mixing.

In light of these observations, we arrive at the final LongPT recipe and train our model, **MMProLong**, with a 5B-token budget. It improves long-document VQA performance by 7.1% at 64K and 128K contexts and maintains strong performance at 256K and 512K without additional training or adaptation, exceeding baselines by over 20%. These gains also transfer to broader multimodal long-context tasks, including webpage-based needle-in-a-haystack on MM-NIAH [17], long-context compression on VTCBench [18], and long-video understanding [7, 8, 19]. Finally, we validate the recipe on Qwen3-VL [3], showing that it is not specific to Qwen2.5-VL and can benefit stronger long-context backbones. Together, these results suggest a practical path toward training long-context vision-language models with data-efficient, transferable LongPT recipes.

2 Related Work

Context window extension. Extending the context window has become a key direction for improving long-context performance, with recent LLMs supporting 128K and even 1M context windows [1, 20–23]. Existing approaches either extend context windows through lightweight methods, such as positional extrapolation [24–28] and attention modifications [29–33], or rely on continued pre-training to build more robust long-context capability [16, 34–36]. Our work follows the continued pre-training method, but studies it in multimodal settings where long contexts contain interleaved image and text tokens.

Long-context vision-language models. As LLM context windows have expanded, recent LVLMs such as Gemini 3.1 Pro [12], Claude Sonnet 4.7 [37], and Qwen3-VL [3] have also supported substantially longer contexts. However, recent LVLM technical reports [3, 14] reveal limited details about how long-context capability is actually built, leaving practical LongPT recipes underexplored. Concurrent work [38] studies long-document data construction for LVLMs, but mainly builds on backbones that already support 128K or longer contexts, such as Qwen3-VL [3] and Mistral 3.1 [39]; thus, its findings may reflect context alignment rather than true context extension. For example, they find that 1B-token LongPT outperforms its 10B-token counterpart, and that LongSFT outperforms LongPT. In contrast, we study LongPT on Qwen2.5-VL [15], whose native context window is only 32K, allowing us to directly examine how to extend LVLMs to longer context. Another line of work studies long-video understanding [40–46], but these methods are often specialized for temporal redundancy and video token reduction rather than general long-context LVLM training.

Multimodal long-context evaluation. Recent benchmarks evaluate multimodal long-context understanding

from diverse perspectives, including long-document VQA [47, 48], multimodal needle-in-a-haystack [6, 49], vision-text compression [18], and long-video understanding [8, 19, 50]. Among them, MMLongBench [5] provides a comprehensive evaluation across five task categories with standardized context lengths up to 128K. Our evaluation covers MMLongBench, VTCBench, and long-video benchmarks, demonstrating the broad generalization of our model **MMPROLONG**.

3 Experimental Setup

We conduct our LongPT experiments using Qwen2.5-VL-7B [15], extending its original 32K context window to 128K. Following the Dynamic-NTK heuristic [51], we scale the mRoPE base frequency from its original value of 1×10^6 to 4×10^6 with detailed ablations provided in Appendix G.3. Each LongPT run is trained with a fixed budget of 5B tokens, a maximum sequence length of 131,072 tokens, and a global batch size of 4M tokens. Throughout the paper, we use binary prefixes: $K = 2^{10}$, $M = 2^{20}$, and $B = 2^{30}$. We provide the full implementation details in Appendix A.2 and the full evaluation details in Appendix B.

4 Multimodal Long-Context Data Curation

Recent studies [14, 52, 53] have identified data synthesis and mixture design as critical factors in pre-training, making data design a central focus of our study. For LVLMS, documents provide a natural source for synthesizing image-text data, as each page combines rich visual layout with dense textual content and can be rendered into long multimodal sequences. In this section, we first describe the preliminary step for constructing the document pool, which provides the raw image-text source for further data synthesis. Next, we discuss five training tasks for synthesizing multimodal long-context data from long documents, grouped into two categories: long-document VQA and OCR transcription. Finally, we conduct experiments to evaluate which task category provides more effective LongPT supervision.

4.1 Document Pool Construction

To support scalable data synthesis, we first construct a large-scale document pool comprising over 1.5 million PDF-formatted documents from multiple sources. The resulting pool spans a broad range of document types, including academic papers, books, and technical manuals, as well as diverse domains such as engineering, medicine, social sciences, and biology. Detailed statistics and domain distribution are provided in Appendix C.1. For data synthesis, we select documents with 32 to 50 pages from this pool. With the 2×2 -pixel unshuffle in our Qwen2.5-VL backbone, these documents yield multimodal sequences ranging from 32K to 128K tokens. To avoid evaluation contamination, we further filter out potential overlap with evaluation benchmarks using SHA-256 hashes of PDF content.

As LVLMS operate on images rather than PDF files, each PDF page is rendered to an image at $DPI = 144$ using PyMuPDF¹. This resolution provides a practical trade-off between visual fidelity and storage cost. In addition, we use an OCR expert model fine-tuned from Seed 2.0 [4] to parse each rendered page into layout-aware blocks. These parsed blocks are further used in both task categories: title and section labels provide the section structure to guide the sampling of coherent page segments for long-document VQA, while recognized text blocks serve as transcription targets for the OCR transcription training tasks.

4.2 Long-Document VQA Data Synthesis

Segment-level synthesis pipeline. We construct the long-document VQA training data using a short-to-long synthesis pipeline. The key idea is to generate a QA pair from a short, semantically coherent page segment, and then place it back into the full-document context to form a long-context training instance.

Specifically, we first parse each document with our OCR expert model and identify its section structure using two element labels, namely **title** and **section**. Based on the parsed structure, we randomly sample one or more consecutive sections whose total length spans 8–15 pages. This produces a coherent page segment at the section level for QA generation.

¹<https://github.com/pymupdf/pymupdf>

Next, we feed the page images of the sampled segment into Seed 2.0 [4], which serves as the QA-generator. We prompt the model to generate a QA pair, along with evidence descriptions and evidence pages, using the detailed prompt provided in [Appendix D.3](#).

Finally, we recover the original full document corresponding to the sampled segment and combine it with the generated QA pair. This yields a single long-context VQA training instance, where the answer can be inferred from a localized short segment while the model must process the full long-document context.

Data quality and efficiency. Since we provide the QA-generation model with only 8–15 pages, the pipeline relies on strong short-context understanding, without requiring full-document processing. In this way, we find that the generated QA pairs are of high quality, and further verify them through a manual check described in [Appendix D.4](#). By sampling short segments, this pipeline is also efficient, substantially reducing the cost of generating large-scale data.

A key challenge in segment-level QA synthesis is ensuring that locally valid questions remain unambiguous when evaluated in the full-document context. Specifically, because QA pairs are generated from a short segment, the same question may have a different answer when placed back into the full document. For example, a question such as “*What is the reported revenue?*” may be answerable within the sampled section, but ambiguous in a full financial report where different sections report revenue for different departments or years. To avoid such global-context false positives, we require the QA-generation model to add explicit segment anchors to the question, such as “*in the Introduction section*” or “*on pages 20–25*”.

Data types. With this segment-level QA synthesis pipeline, we synthesize three training tasks of long-document VQA data, each targeting a distinct capability defined by the type and number of evidence pieces required to answer the question. They cover increasing evidence complexity: (i) **single-page extraction** (`extract-single`) asks the model to retrieve factual information from a single page, e.g., “According to the Homemade Bitters recipe on Page 39, how long should the herbs soak in vodka?”; (ii) **multi-page extraction** (`extract-multi`) requires the model to aggregate factual information from multiple pages, e.g., “Based on Pages 6, 13, and 19, list all risk factors mentioned in the report.”; and (iii) **reasoning** (`reasoning`) further requires numerical or logical operations over extracted information, such as summation, comparison, or counting across pages, e.g., “What is the difference between total consumption and total imports for rice production in 2020?”.

Together, the first two training tasks focus on locating and extracting relevant evidence from long documents, while the reasoning task further evaluates whether the model can operate on the extracted evidence.

4.3 OCR Transcription Data Synthesis

In addition to long-document VQA, another category of long-context training tasks we build is OCR transcription. This task category encourages LVLMs to capture long-distance image-text dependencies by requiring them to transcribe text elements across all pages of a long document.

Synthesis pipeline. For each document, we first parse every page with our OCR expert and retain text elements such as section titles, paragraphs, tables, and captions. We then construct an OCR transcription sequence by using the rendered page images as visual input and the parsed text elements as the target output.

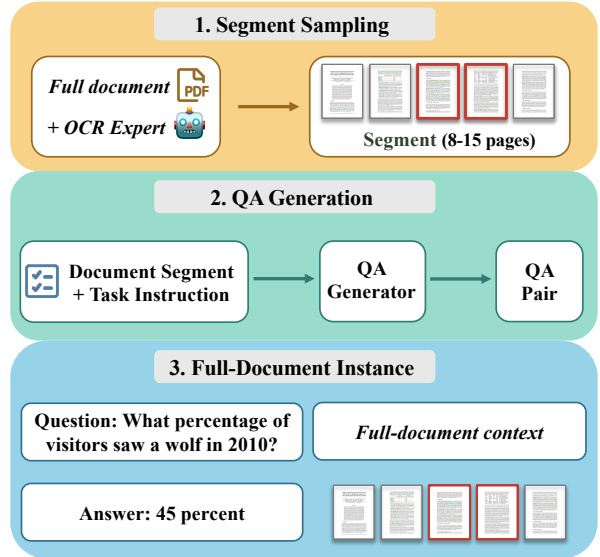


Figure 1 Long-document VQA synthesis pipeline. We first parse the full document with an OCR expert and sample a coherent segment. An LVLm used as a QA generator (e.g., Seed 2.0) then produces a QA pair from the segment, which is inserted back into the original document to form a long-context training instance.

Table 1 We compare long-document VQA data with OCR transcription data under the same setting. The models are evaluated on the document category of MMLongBench [5] at 64K and 128K, which contains three datasets: MMLongBench-Doc [47], LongDocURL [48], and SlideVQA [55]. We abbreviate them as MMLB-D, LD-URL, and SLIDE, respectively. **SFT** means an extra 5B-token SFT stage.

Training data	64K MMLongBench				128K MMLongBench				AVG.
	MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
Qwen2.5-VL-7B	32.17	49.57	75.00	52.24	26.96	51.85	68.00	48.94	50.59
extract-single	33.85	59.73	77.00	56.86	30.89	55.69	77.00	54.53	55.69 ^{+5.1}
extract-multi	32.75	64.32	77.00	58.02	31.50	54.82	81.00	55.77	56.90 ^{+6.3}
reasoning	32.67	60.34	79.00	57.33	29.23	61.61	76.00	55.62	56.47 ^{+5.9}
OCR-full	23.67	11.06	59.00	31.24	23.97	20.36	61.00	35.11	33.17 ^{-17.4}
OCR-needle	40.07	28.77	68.00	45.61	22.75	37.24	66.00	42.00	43.80 ^{-6.8}
OCR-full (SFT)	37.19	53.07	78.00	56.09	25.03	51.73	78.00	51.59	53.84 ^{+3.2}
OCR-needle (SFT)	33.79	50.38	78.00	54.06	26.65	49.84	76.00	50.83	52.44 ^{+1.9}

With this formulation, LVLMS must repeatedly attend back to the dense textual content in the rendered images and transcribe it over long distances, thereby modeling long-distance image-text dependencies.

Data types. Using this pipeline, we generate two training tasks of OCR transcription data. These types are defined by the scope of pages to be transcribed: (i) **full-document OCR** (OCR-full) requires the model to transcribe text elements from all pages of the document, encouraging dense image-text dependency modeling across the full context; and (ii) **needle-page OCR** (OCR-needle) selects only a small subset of pages (1–3 pages) for transcription and keeps the remaining pages as distractors, turning OCR transcription into a retrieval-style long-context training task. Collectively, these two tasks encourage LVLMS to model long-distance image-text dependencies under both dense transcription and retrieval-style settings.

4.4 Comparing Long-Document VQA and OCR Transcription

We compare the five candidate tasks under a controlled 5B-token budget. For each task, we build a separate training set and train Qwen2.5-VL-7B [15] using the hyperparameters in Section 3. Dataset statistics, such as token counts and sequence-length distributions, are provided in Appendices D.1 and E.1.

SFT after LongPT on OCR Transcription. OCR transcription encourages long-distance image-text dependency modeling but is not naturally aligned with instruction-following evaluations. To favor OCR-based LongPT, we further apply a 5B-token SFT stage to the OCR-trained checkpoints using LLaVA-OneVision instruction data [54]; data details are in Appendix F.

Long-document VQA provides stronger supervision. The results are shown in Table 1. First, the 32K base model degrades substantially at 128K, with MMLongBench-Doc dropping from 32.17% to 26.96%. More importantly, OCR transcription tasks yield poor downstream performance, especially full-document OCR, whose overall average drops by 17.4% to 33.17%. After adding the SFT stage to improve instruction-following ability, the OCR-trained checkpoints obtain moderate gains of 3.24% and 1.85% for full-document and needle-page OCR, respectively. In contrast, all three long-document VQA tasks consistently improve performance by more than 5% in absolute terms, with multi-page extraction achieving the best average of 56.90%. This makes long-document VQA a stronger and more computationally efficient supervision source for LongPT, yielding better downstream performance without an additional 5B-token SFT stage. Its advantage suggests that instruction-formatted supervision and task diversity, ranging from information extraction to complex numerical reasoning, are important for LongPT. We therefore focus on long-document VQA in the remaining data-design experiments.

5 Data Mixture and Training Design

Having identified long-document VQA as an effective data source, we now study how to turn it into a practical LongPT recipe. Specifically, we examine three key design choices: the distribution of training instance lengths,

Table 2 Long-context data mixture test. We grid search the mix of information extraction vs. reasoning tasks from 0 : 10 to 10 : 0 under the fixed 5B-token budget. **Ratio** represents the extraction-to-reasoning ratio.

Ratio	64K MMLongBench				128K MMLongBench				AVG.
	MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
0:10	32.67	60.34	79.00	57.33	29.23	61.61	76.00	55.62	56.47
2:8	32.04	65.02	77.00	58.02	27.37	61.34	74.00	54.24	56.13
4:6	31.57	59.49	78.00	56.35	32.95	53.39	79.00	55.11	55.73
6:4	32.52	64.86	79.00	58.79	32.72	55.53	79.00	55.75	57.27
8:2	36.00	62.69	80.00	59.56	34.19	56.33	77.00	55.84	57.70
10:0	33.98	59.48	79.00	57.49	32.12	59.07	78.00	56.40	56.94

the mixture of long-context data, and the preservation of short-context performance. We provide an additional ablation on the RoPE base frequency in [Appendix G.3](#).

5.1 Training Sequence-Length Distribution

When extending the context window of LLMs, prior work [16, 35] often relies on books or code repositories from SlimPajama [56] or the Stack [57], whose sequence lengths are naturally distributed across 8K to 128K tokens. In contrast, our long-document pool contains a large number of documents ranging from 20 to 200 pages, providing sufficient coverage for constructing training instances at different target lengths. This raises a practical question: how should we choose the length distribution of synthesized training instances?

Constructing data with different length distributions.

Here, we study two length distributions, namely *pool-native* and *long-biased*. In the data-curation study ([Section 4](#)), we use the *pool-native* length distribution by default, as training instances are synthesized from documents naturally sampled within the 32–50 page range, without additional length-based reweighting.

Given that we evaluate our models at a 128K context length, it is natural to ask whether allocating the token budget to longer examples leads to better LongPT results. We therefore construct the *long-biased* variant of data in which 83.9% of the examples contain at least 100K tokens, compared with only 23.6% in the *pool-native* distribution (See [Table 11](#)). This variant exposes the model more frequently to near-maximum-length contexts (128K), whereas the pool-native distribution covers a broader range of context lengths. Detailed statistics for both distributions are summarized in [Appendices D.1](#) and [D.2](#).

A diverse length distribution yields better long-context capability. The average performance of both length distributions is summarized in [Figure 2](#), with full evaluation results provided in [Appendix G.1](#). Overall, the *pool-native* distribution outperforms the *long-biased* distribution, yielding average gains of +1.3, +0.1, and +1.7 points for **extract-single**, **extract-multi**, and **reasoning** tasks, respectively. Notably, the *pool-native* distribution consistently matches or outperforms the *long-biased* distribution while containing fewer near-128K training examples.

These empirical findings suggest that long-context ability is not a discrete capability acquired only at a specific target length, such as 128K context. Instead, it requires continuous calibration across different absolute positions and relative image-text distances. In other words, LongPT should teach the model to retrieve key information in a way that generalizes across diverse long-context scenarios. Based on this observation, we adopt the *pool-native* distribution for all the following experiments in this paper.

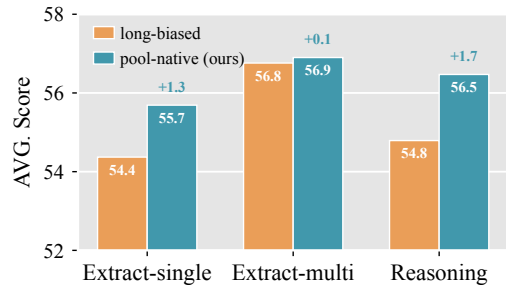


Figure 2 Comparison between different length distributions. We report overall average scores across 64K and 128K of the long-document VQA from MMLongBench. See full results in [Appendix G.1](#).

Table 3 Short-context performance under different short-data mixing ratios. We report the average over six short-context benchmarks together with per-benchmark scores. **Short Data** means the proportion of short-context data. 0% means only using long-context data.

Short Data	General VQA		Multimodal Reasoning			Text Rec.	AVG.
	MMBench	RWQA	MMMU	MMMU-Pro	MathVista	OCRBench	
Qwen2.5-VL-7B	80.68	68.76	53.00	37.80	70.50	88.10	66.47
0%	80.00	72.68	49.33	36.65	68.70	85.50	65.48
20%	81.82	70.98	54.33	36.47	68.30	87.30	66.53
40%	81.25	70.46	52.67	37.05	68.30	87.10	66.14
60%	81.66	69.93	52.67	37.51	67.80	86.70	66.05
80%	81.14	69.67	52.11	37.40	69.30	87.40	66.17

5.2 Multi-Task Long-Context Data Mixture

Data composition is critical for long-context extension [16, 35]. So far, our studies have evaluated each training task of long-document VQA data in isolation. We therefore study how to combine these three types of data into a single training mixture. Specifically, we group them into two categories: information extraction, which combines `extract-single` and `extract-multi` evenly, and reasoning, which corresponds to `reasoning`.

We grid search the extraction-to-reasoning ratio in 20% increments, ranging from all reasoning (0:10) to all extraction (10:0). As shown in Table 2, moderately extraction-heavy mixtures perform best, with 6:4 and 8:2 achieving the highest overall scores. The best performance is obtained with an extraction-to-reasoning ratio of 8:2, which also outperforms the best single-data setting in Table 1. This suggests that combining complementary training tasks is more effective than relying on any single task alone. It also indicates that retrieving key information from long-context inputs remains a major bottleneck when extending the context window, while retaining a small amount of reasoning data helps preserve task diversity. We therefore use the 8:2 mixture in subsequent experiments.

5.3 Short-Context Performance Preservation

The degradation of short-context capabilities is a common concern in long-context continued pre-training. To examine this trade-off, we mix different proportions of short-context data into the LongPT stage while keeping the total token budget fixed. We obtain the short-context data from LLaVA-OneVision [54], with details provided in Appendix F. We keep the 8:2 extraction-to-reasoning mixture fixed as the long-context component and vary the short-context data ratio from 0% to 80% in increments of 20%. We summarize the averages of long-context performance in Figure 3, with full results provided in Appendix G.2. In particular, we report short-context performance on six benchmarks across three capabilities in Table 3, with evaluation details provided in Appendix B.7.

Results and trade-off. A somewhat surprising observation is that pure long-context training does not severely degrade short-context capabilities. With 0% short data, the model achieves the best long-document VQA average of 57.70, with only a mild drop in short-context average from 66.47 to 65.48. This suggests that high-quality long-document VQA data can preserve the model’s general short-context ability, possibly because its QA format still follows an instruction-following style despite the substantially longer input context.

Meanwhile, short-context mixing introduces a clear trade-off. Adding 20% short-context data yields the best short-context average of 66.53, but lowers the long-document VQA score to 55.57. In contrast, the 40% setting

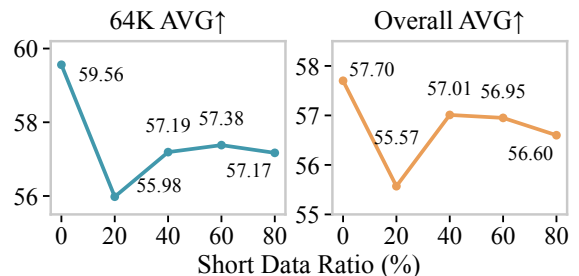


Figure 3 Long-document VQA performance under different short-data mixing ratios. We report 64K AVG and the overall AVG across 64K and 128K. Full results, including 128K scores, are provided in Appendix G.2.

Table 4 Final long-document VQA results. We compare **MMPProLong**, trained with the final LongPT recipe, against representative open-source and closed-source LVLMs at 64K and 128K.

Model	64K MMLongBench				128K MMLongBench				AVG.
	MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
Small open-source LVLMs (< 15B)									
MMPProLong (ours)	36.00	62.69	80.00	59.56	34.19	56.33	77.00	55.84	57.70
Qwen2.5-VL-7B	32.17	49.57	75.00	52.24	26.96	51.85	68.00	48.94	50.59
InternVL3-8B	27.72	52.84	70.00	50.19	25.89	48.44	58.00	44.11	47.15
InternVL3-14B	35.33	49.69	67.00	50.67	29.61	50.21	53.00	44.27	47.47
InternVL3.5-8B	29.26	47.94	38.00	38.40	18.28	31.89	0.00	16.72	27.56
InternVL3.5-14B	29.04	48.57	53.00	43.54	16.91	33.86	0.00	16.92	30.23
Gemma3-4B	25.00	33.33	57.00	38.44	26.44	33.38	53.00	37.61	38.03
Gemma3-12B	31.52	49.59	63.00	48.03	30.56	51.91	60.00	47.49	47.76
Gemma4-E2B	28.00	29.86	48.00	35.29	20.83	27.73	51.00	33.19	34.24
Gemma4-E4B	36.00	30.21	45.00	37.07	28.50	37.78	45.00	37.09	37.08
Large open-source LVLMs (≥ 15B)									
Qwen2.5-VL-32B	44.06	60.58	77.00	60.55	32.08	60.15	76.00	56.08	58.31
Qwen2.5-VL-72B	51.97	58.96	77.5	62.81	40.19	62.45	73.9	58.85	60.83
InternVL3-38B	36.00	49.32	74.00	53.11	29.65	50.33	54.00	44.66	48.88
InternVL3.5-38B	39.17	48.37	52.00	46.51	22.24	40.09	1.01	21.11	33.81
Gemma3-27B	38.00	51.72	69.00	52.91	27.94	61.13	68.00	52.35	52.63
Gemma4-26B-A4B	40.67	42.82	64.00	49.16	35.68	43.25	60.00	46.31	47.74
Gemma4-31B	56.33	62.42	84.00	67.58	51.87	62.00	86.00	66.62	67.10
Closed-source LVLMs									
GPT-5.4	61.56	72.37	87.10	73.68	52.96	73.06	N/A	63.01	69.41
GPT-5.5	93.10	92.52	95.77	93.80	83.33	89.12	N/A	86.22	90.77
Gemini-2.5-Pro	65.40	68.13	89.00	74.18	65.04	67.60	91.00	74.55	74.37
Gemini-3.1-Flash	74.43	79.32	91.00	81.58	69.63	76.38	90.00	78.67	80.13
Gemini-3.1-Pro	79.40	78.25	93.00	83.55	77.75	80.57	93.00	83.77	83.66

provides a better practical balance: it better preserves long-document performance, averaging 57.01, while keeping short-context performance close to the original model at 66.14.

Overall, since our goal is to maximize long-context capability without substantial degradation in short-context performance, we adopt pure long-context training without short-context data in the final recipe. The 40% setting can be viewed as a balanced alternative when stronger short-context preservation is required.

6 MMPProLong Performance and Generalization

Based on the above design choices, we arrive at the final training recipe of **MMPProLong**, with the full configuration in [Appendix A.1](#). In this section, we first compare **MMPProLong** with a wide range of LVLm baselines on long-document VQA. Besides this, we further show that **MMPProLong** can generalize to (i) longer contexts up to 512K without further training or adaptation; (ii) diverse long-context tasks without task-specific training, including MM-NIAH for webpage-based needle-in-a-haystack evaluation, VTCBench for long-context vision-text compression, and long-video understanding benchmarks. To further evaluate the generalizability of our training recipe, we also validate it on Qwen3-VL-8B, with results shown in [Appendix G.7](#).

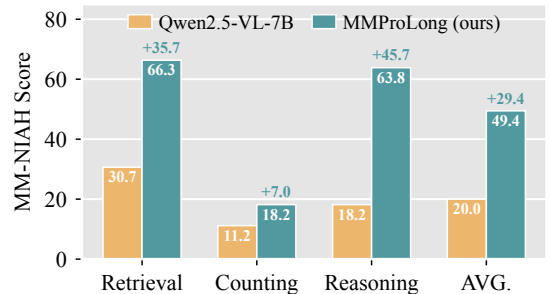


Figure 4 MM-NIAH Scores. We report scores averaged over 64K and 128K contexts for retrieval, counting, reasoning, and the overall average. Additional baselines are provided in [Appendix G.4](#).

Table 5 We evaluate the final 128K **MMPROLONG** at 256K and 512K without additional training or adaptation. MMLongBench-Doc (MMLB-D), LongDocURL (LD-URL), and SlideVQA (SLIDE) are reported in order. Some baselines fail on 512K SlideVQA due to the large number of images.

Model	256K MMLongBench				512K MMLongBench				AVG.
	MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
MMPROLONG	29.69	58.58	77.00	55.09	31.91	55.65	70.00	52.52	53.80
Qwen2.5-VL-7B	25.47	35.88	53.00	38.12	13.44	24.61	20.41	19.49	28.80
Gemma3-4B	20.89	32.68	44.00	32.52	20.39	26.14	0.00	15.51	24.02
Gemma3-12B	31.63	47.47	63.00	47.37	24.18	46.37	0.00	23.51	35.44

6.1 MMPROLONG Compared with LVLM Baselines

We compare **MMPROLONG**, trained with the final recipe, against a wide range of open-source and closed-source LVLM baselines, as shown in Table 4. The full list of evaluated models is provided in Appendix B.2. Among open-source LVLMs below 15B parameters, **MMPROLONG** achieves the best overall average of 57.70, improving the Qwen2.5-VL-7B base model from 50.59 by 7.11%. The improvement is consistent across both context lengths, raising the average score from 52.24 to 59.56 at 64K and from 48.94 to 55.84 at 128K. Notably, **MMPROLONG** also outperforms several substantially larger open-source LVLMs, including InternVL3-38B and Gemma3-27B, achieving competitive long-context performance among open-source models.

6.2 Generalization Beyond the Training Setting

Generalization to longer contexts. In this experiment, we examine whether the final **MMPROLONG** recipe can extrapolate beyond its 128K training context. To this end, we further extend the long-document VQA benchmarks to 256K and 512K contexts, with details of context extension described in Appendix B.3. As shown in Table 5, **MMPROLONG** generalizes to longer contexts without additional training or adaptation. While Qwen2.5-VL-7B degrades sharply as the context length increases, dropping from 38.12 at 256K to 19.49 at 512K, **MMPROLONG** maintains strong performance at both lengths, achieving 55.09 at 256K and 52.52 at 512K. This results in a significant overall advantage over the original Qwen2.5-VL-7B model, increasing the average score from 28.80 to 53.80.

Having shown that the final recipe improves long-document VQA performance and extrapolates to longer contexts, we next examine whether the learned long-context capability transfers to other multimodal tasks. Figures 4 and 5 report results on MM-NIAH and long-video understanding benchmarks, namely Video-MME [8], MLVU [19], and LongVideoBench [7]. We also evaluate long-context vision-text compression on VTCBench [18], with full results provided in Appendix G.5.

MM-NIAH [6] is a multimodal needle-in-a-haystack benchmark that evaluates retrieval, counting, and reasoning tasks over webpage-based haystacks. As shown in Figure 4, **MMPROLONG** substantially improves over Qwen2.5-VL-7B, increasing the average score from 20.0 to 49.4. The gains are especially pronounced on retrieval and reasoning tasks, suggesting that long-document VQA training improves the model’s ability to locate sparse evidence and use it for reasoning in long multimodal contexts.

We observe similar transfer on long-video understanding benchmarks and VTCBench. On long-video benchmarks in Figure 5, **MMPROLONG** consistently improves over Qwen2.5-VL-7B on Video-MME, MLVU, and LongVideoBench, despite not using video-specific training data; detailed scores are provided in Appendix G.6. Meanwhile, VTCBench evaluates LVLMs’ ability to perform long-context vision-text compression across three tasks: retrieval, reasoning, and memory. As shown in Table 18 in Appendix G.5, **MMPROLONG** improves the overall score on VTCBench from 48.23 to 52.73, with gains on both reasoning and memory tasks while main-

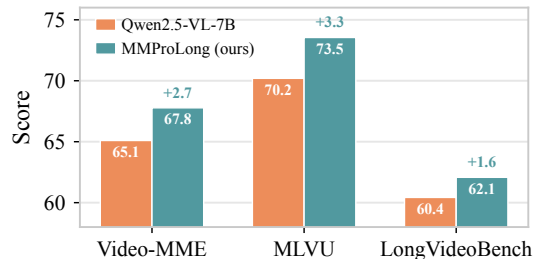


Figure 5 Long-video generalization. We report scores on Video-MME, MLVU, and LongVideoBench. Full results are provided in Appendix G.6.

taining strong retrieval performance. Together, these results indicate that our proposed LongPT recipe learns a general long-context multimodal capability rather than overfitting to the document VQA format.

7 Conclusion

In this work, we presented a systematic study of long-context continued pre-training for LVLMs, focusing on how to construct and mix effective multimodal long-context data. Our experiments show that long-document VQA is a strong and practical training task, as it provides diverse retrieval and reasoning supervision signals, preserves short-context capability, and supports context extension from 32K to 128K under a modest token budget. Instantiated as **MMProLong**, our recipe improves long-document VQA performance and generalizes beyond the training context window to 256K and 512K context lengths, as well as broader multimodal long-context tasks, such as long-video understanding. We hope our study provides a practical foundation for building future LVLMs with reliable long-context capability.

References

- [1] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [2] Meta. The llama 4 herd: The beginning of a new era of natively multimodal ai innovation, 2025. URL <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>, 2025.
- [3] Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- [4] Bytedance Seed. Seed2. 0 model card: Towards intelligence frontier for real-world complexity. Technical report, Technical report, Bytedance, 2025. URL <https://lf3-static.bytednsdoc.com/...>
- [5] Zhaowei Wang, Wenhao Yu, Xiyu Ren, Jipeng Zhang, Yu Zhao, Rohit Saxena, Liang Cheng, Ginny Wong, Simon See, Pasquale Minervini, et al. Mmlongbench: Benchmarking long-context vision-language models effectively and thoroughly. *arXiv preprint arXiv:2505.10610*, 2025.
- [6] Hengyi Wang, Haizhou Shi, Shiwei Tan, Weiyei Qin, Wenyuan Wang, Tunyu Zhang, Akshay Nambi, Tanuja Ganu, and Hao Wang. Multimodal needle in a haystack: Benchmarking long-context capability of multimodal large language models. *arXiv preprint arXiv:2406.11230*, 2024.
- [7] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2024.
- [8] Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24108–24118, 2025.
- [9] Xinyu Geng, Peng Xia, Zhen Zhang, Xinyu Wang, Qiuchen Wang, Ruixue Ding, Chenxi Wang, Jialong Wu, Yida Zhao, Kuan Li, et al. Webwatcher: Breaking new frontier of vision-language deep research agent. *arXiv preprint arXiv:2508.05748*, 2025.
- [10] Huanyao Zhang, Jiepeng Zhou, Bo Li, Bowen Zhou, Yanzhe Shan, Haishan Lu, Zhiyong Cao, Jiaoyang Chen, Yuqian Han, Zinan Sheng, et al. Browsecomp-v3: A visual, vertical, and verifiable benchmark for multimodal browsing agents. *arXiv preprint arXiv:2602.12876*, 2026.
- [11] Yuxuan Zhang, Yubo Wang, Yipeng Zhu, Penghui Du, Junwen Miao, Xuan Lu, Wendong Xu, Yunzhuo Hao, Songcheng Cai, Xiaochen Wang, et al. Clawbench: Can ai agents complete everyday online tasks? *arXiv preprint arXiv:2604.08523*, 2026.
- [12] The Gemini Team. Gemini 3.1 pro: A smarter model for your most complex tasks. <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/>, Feb 2026. Google The Keyword Blog.
- [13] OpenAI. Introducing GPT-5.4. <https://openai.com/index/introducing-gpt-5-4/>, Mar 2026. OpenAI Blog.
- [14] Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, Lihang Pan, et al. Glm-4.5 v and glm-4.1 v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning. *arXiv preprint arXiv:2507.01006*, 2025.
- [15] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- [16] Tianyu Gao, Alexander Wettig, Howard Yen, and Danqi Chen. How to train long-context language models (effectively). *arXiv preprint arXiv:2410.02660*, 2024.
- [17] Weiyun Wang, Shuibo Zhang, Yiming Ren, Yuchen Duan, Tiantong Li, Shuo Liu, Mengkang Hu, Zhe Chen, Kaipeng Zhang, Lewei Lu, et al. Needle in a multimodal haystack. *Advances in Neural Information Processing Systems*, 37:20540–20565, 2024.
- [18] Hongbo Zhao, Meng Wang, Fei Zhu, Wenzhuo Liu, Bolin Ni, Fanhu Zeng, Gaofeng Meng, and Zhaoxiang Zhang. Vtcbench: Can vision-language models understand long context with vision-text compression? *arXiv preprint arXiv:2512.15649*, 2025.

- [19] Junjie Zhou, Yan Shu, Bo Zhao, Boya Wu, Zhengyang Liang, Shitao Xiao, Minghao Qin, Xi Yang, Yongping Xiong, Bo Zhang, et al. Mlvu: Benchmarking multi-task long video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13691–13701, 2025.
- [20] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- [21] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- [22] Google. Introducing gemini 2.0: our new ai model for the agentic era, 2024. URL <https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/#ceo-message>.
- [23] Google. Gemini 2.5: Our most intelligent ai model. URL <https://blog.google/technology/google-deepmind/gemini-model-thinking-updates-march-2025/>, 2025.
- [24] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023.
- [25] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. Yarn: Efficient context window extension of large language models. In *The Twelfth International Conference on Learning Representations*.
- [26] Yiran Ding, Li Lina Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. Longrope: Extending llm context window beyond 2 million tokens. *arXiv preprint arXiv:2402.13753*, 2024.
- [27] Yikai Zhang, Junlong Li, and Pengfei Liu. Extending llms’ context window with 100 samples. *arXiv preprint arXiv:2401.07004*, 2024.
- [28] Dawei Zhu, Nan Yang, Liang Wang, Yifan Song, Wenhao Wu, Furu Wei, and Sujian Li. Pose: Efficient context window extension of llms via positional skip-wise training. *arXiv preprint arXiv:2309.10400*, 2023.
- [29] Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023.
- [30] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023.
- [31] Chaojun Xiao, Pengl Zhang, Xu Han, Guangxuan Xiao, Yankai Lin, Zhengyan Zhang, Zhiyuan Liu, Song Han, and Maosong Sun. Infilmm: Unveiling the intrinsic capacity of llms for understanding extremely long sequences with training-free memory. *arXiv preprint arXiv:2402.04617*, 3(7), 2024.
- [32] Amanda Bertsch, Uri Alon, Graham Neubig, and Matthew Gormley. Unlimiformer: Long-range transformers with unlimited length input. *Advances in Neural Information Processing Systems*, 36:35522–35543, 2023.
- [33] Hongye Jin, Xiaotian Han, Jingfeng Yang, Zhimeng Jiang, Zirui Liu, Chia-Yuan Chang, Huiyuan Chen, and Xia Hu. Llm maybe longlm: Self-extend llm context window without tuning. *arXiv preprint arXiv:2401.01325*, 2024.
- [34] Wenhan Xiong, Jingyu Liu, Igor Molybog, Hejia Zhang, Prajjwal Bhargava, Rui Hou, Louis Martin, Rashi Rungta, Karthik Abinav Sankararaman, Barlas Oguz, et al. Effective long-context scaling of foundation models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4643–4663, 2024.
- [35] Yao Fu, Rameswar Panda, Xinyao Niu, Xiang Yue, Hannaneh Hajishirzi, Yoon Kim, and Hao Peng. Data engineering for scaling language models to 128K context. *arXiv preprint arXiv:2402.10171*, 2024.
- [36] An Yang, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoyang Huang, Jiandong Jiang, Jianhong Tu, Jianwei Zhang, Jingren Zhou, Junyang Lin, Kai Dang, Kexin Yang, Le Yu, Mei Li, Minmin Sun, Qin Zhu, Rui Men, Tao He, Weijia Xu, Wenbiao Yin, Wenyuan Yu, Xiafei Qiu, Xingzhang Ren, Xinlong Yang, Yong Li, Zhiying Xu, and Zipeng Zhang. Qwen2.5-1m technical report. *ArXiv*, abs/2501.15383, 2025. URL <https://api.semanticscholar.org/CorpusID:275921951>.
- [37] Anthropic. Introducing Claude Opus 4.7. <https://www.anthropic.com/news/claude-opus-4-7>, Apr 2026. Anthropic News.
- [38] Austin Veselka. How to train your long-context visual document model. *arXiv preprint arXiv:2602.15257*, 2026.

- [39] Mistral AI. Mistral small 3.1. <https://mistral.ai/news/mistral-small-3-1/>, Mar 2025. Mistral AI Research.
- [40] Yukang Chen, Fuzhao Xue, Dacheng Li, Qinghao Hu, Ligeng Zhu, Xiuyu Li, Yunhao Fang, Haotian Tang, Shang Yang, Zhijian Liu, et al. Longvila: Scaling long-context visual language models for long videos. *arXiv preprint arXiv:2408.10188*, 2024.
- [41] Yunhang Shen, Chaoyou Fu, Shaoqi Dong, Xiong Wang, Yi-Fan Zhang, Peixian Chen, Mengdan Zhang, Haoyu Cao, Ke Li, Shaohui Lin, et al. Long-vita: Scaling large multi-modal models to 1 million tokens with leading short-context accuracy. *arXiv preprint arXiv:2502.05177*, 2025.
- [42] Te Yang, Xiangyu Zhu, Bo Wang, Quan Chen, Peng Jiang, and Zhen Lei. Eea: Exploration-exploitation agent for long video understanding. *arXiv preprint arXiv:2512.03500*, 2025.
- [43] Peiyuan Zhang, Kaichen Zhang, Bo Li, Guangtao Zeng, Jingkan Yang, Yuanhan Zhang, Ziyue Wang, Haoran Tan, Chunyuan Li, and Ziwei Liu. Long context transfer from language to vision. *arXiv preprint arXiv:2406.16852*, 2024.
- [44] Zuhao Yang, Sudong Wang, Kaichen Zhang, Keming Wu, Sicong Leng, Yifan Zhang, Bo Li, Chengwei Qin, Shijian Lu, Xingxuan Li, et al. Longvt: Incentivizing "thinking with long videos" via native tool calling. *arXiv preprint arXiv:2511.20785*, 2025.
- [45] Shuming Liu, Chen Zhao, Tianqi Xu, and Bernard Ghanem. Bolt: Boost large vision-language model without training for long-form video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 3318–3327, 2025.
- [46] Xiaoqian Shen, Yunyang Xiong, Changsheng Zhao, Lemeng Wu, Jun Chen, Chenchen Zhu, Zechun Liu, Fanyi Xiao, Balakrishnan Varadarajan, Florian Bordes, et al. Longvu: Spatiotemporal adaptive compression for long video-language understanding. *arXiv preprint arXiv:2410.17434*, 2024.
- [47] Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, et al. Mmlongbench-doc: Benchmarking long-context document understanding with visualizations. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- [48] Chao Deng, Jiale Yuan, Pi Bu, Peijie Wang, Zhong-Zhi Li, Jian Xu, Xiao-Hui Li, Yuan Gao, Jun Song, Bo Zheng, et al. Longdocurl: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1135–1159, 2025.
- [49] Tsung-Han Wu, Giscard Biamby, Jerome Quenum, Ritwik Gupta, Joseph E Gonzalez, Trevor Darrell, and David M Chan. Visual haystacks: A vision-centric needle-in-a-haystack benchmark. *arXiv preprint arXiv:2407.13766*, 2024.
- [50] Weihang Wang, Zehai He, Wenyi Hong, Yean Cheng, Xiaohan Zhang, Ji Qi, Ming Ding, Xiaotao Gu, Shiyu Huang, Bin Xu, et al. Lvbench: An extreme long video understanding benchmark. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 22958–22967, 2025.
- [51] Emozilla. Dynamically scaled rope further increases performance of long context llama with zero fine-tuning, 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/14mrgpr/dynamically_scaled_rope_further_increases/.
- [52] Gemma Team Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ram'e, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean-Bastien Grill, Sabella Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, Gael Liu, Francesco Visin, Kathleen Kenealy, Lucas Beyer, Xiaohai Zhai, Anton Tsitsulin, Róbert Istvan Busa-Fekete, Alex Feng, Naveen Sachdeva, Benjamin Coleman, Yi Gao, Basil Mustafa, Iain Barr, Emilio Parisotto, David Tian, Matan Eyal, Colin Cherry, Jan-Thorsten Peter, Danila Sinopalnikov, Surya Bhupatiraju, Rishabh Agarwal, Mehran Kazemi, Dan Malkin, Ravin Kumar, David Vilar, Idan Brusilovsky, Jiaming Luo, Andreas Steiner, Abe Friesen, Abhanshu Sharma, Abheesh Sharma, Adi Mayrav Gilady, Adrian Goedeckemeyer, Alaa Saade, Alexander Kolesnikov, Alexei Bendebury, Alvin Abdagig, Amit Vadi, Andr'as Gyorgy, André Susano Pinto, Anil Das, Ankur Bapna, Antoine Miech, Antoine Yang, Antonia Paterson, Ashish Shenoy, Ayan Chakrabarti, Bilal Piot, Boxi Wu, Bobak Shahriari, Bryce Petrini, Charlie Chen, Charline Le Lan, Christopher A. Choquette-Choo, CJ Carey, Cormac Brick, Daniel Deutsch, Danielle Eisenbud, Dee Cattle, Derek Cheng, Dimitris Paparas, Divyashree Shivakumar Sreepathihalli, Doug Reid, Dustin Tran, Dustin Zelle, Eric Noland, Erwin Huizenga, Eugene Kharitonov, Frederick Liu, Gagik Amirkhanyan, Glenn Cameron, Hadi Hashemi, Hanna Klimczak-Plucińska, Harman Singh, Harsh Mehta, Harshal Tushar Lehri, Hussein Hazimeh, Ian Ballantyne, Idan Szepktor,

- Ivan Nardini, Jean Pouget-Abadie, Jetha Chan, Joe Stanton, J. Michael Wieting, Jonathan Lai, Jordi Orbay, Joe Fernandez, Joshua Newlan, Junsong Ji, Jyotinder Singh, Kat Black, Kathy Yu, Kevin Hui, Kiran Vodrahalli, Klaus Greff, Linhai Qiu, Marcella Valentine, Marina Coelho, Marvin Ritter, Matt Hoffman, Matthew Watson, Mayank Chaturvedi, Michael Moynihan, Min Ma, Nabila Babar, Natasha Noy, Nathan Byrd, Nick Roy, Nikola Momchev, Nilay Chauhan, Oskar Bunyan, Pankil Botarda, Paul Caron, Paul Kishan Rubenstein, Phil Culliton, Philipp Schmid, Pier Giuseppe Sessa, Pingmei Xu, Piotr Stańczyk, Pouya Dehghani Tafti, Rakesh Shivanna, Renjie Wu, Renke Pan, Reza Ardeshtir Rokni, Rob Willoughby, Rohith Vallu, Ryan Mullins, Sammy Jerome, Sara Smoot, Sertan Girgin, Shariq Iqbal, Shashir Reddy, Shruti Sheth, Siim Pöder, Sijal Bhatnagar, Sindhu Raghuram Panyam, Sivan Eiger, Susan Zhang, Tianqi Liu, Trevor Yacovone, Tyler Liechty, Uday Kalra, Utku Evci, Vedant Misra, Vincent Roseberry, Vladimir Feinberg, Vlad Kolesnikov, Woohyun Han, Woosuk Kwon, Xi Chen, Yinlam Chow, Yuvein Zhu, Zichuan Wei, Zoltan Egged, Victor Cotruta, Minh Giang, Phoebe Kirk, Anand Rao, Jessica Lo, Erica Moreira, Luiz Gustavo Martins, Omar Sanseviero, Lucas Gonzalez, Zach Gleicher, Tris Warkentin, Vahab S. Mirrokni, Evan Senter, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, Yossi Matias, D. Sculley, Slav Petrov, Noah Fiedel, Noam M. Shazeer, Oriol Vinyals, Jeffrey Dean, Demis Hassabis, Koray Kavukcuoglu, Clément Farabet, Elena Buchatskaya, Jean-Baptiste Alayrac, Rohan Anil, Dmitry Lepikhin, Sebastian Borgeaud, Olivier Bachem, Armand Joulin, Alek Andreev, Cassidy Hardin, Robert Dadashi, and L’eonard Hussenot. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- [53] Google DeepMind. Gemma 4: Our most intelligent open models. *URL* <https://deepmind.google/models/gemma/gemma-4/>, 2026.
- [54] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024.
- [55] Ryota Tanaka, Kyosuke Nishida, Kosuke Nishida, Taku Hasegawa, Itsumi Saito, and Kuniko Saito. Slidevqa: A dataset for document visual question answering on multiple images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 13636–13645, 2023.
- [56] Daria Soboleva, Faisal Al-Khateeb, Robert Myers, Jacob R Steeves, Joel Hestness, and Nolan Dey. Slimpajama: A 627b token cleaned and deduplicated version of redpajama, 2023.
- [57] Denis Kocetkov, Raymond Li, Loubna Ben Allal, Jia Li, Chenghao Mou, Carlos Muñoz Ferrandis, Yacine Jernite, Margaret Mitchell, Sean Hughes, Thomas Wolf, et al. The stack: 3 tb of permissively licensed source code. *arXiv preprint arXiv:2211.15533*, 2022.
- [58] Qianli Ma, Yaowei Zheng, Zhelun Shi, Zhongkai Zhao, Bin Jia, Ziyue Huang, Zhiqi Lin, Youjie Li, Jiacheng Yang, Yanghua Peng, et al. Veomni: Scaling any modality model training with model-centric distributed recipe zoo. *arXiv preprint arXiv:2508.02317*, 2025.
- [59] Tri Dao. Flashattention-2: Faster attention with better parallelism and work partitioning. In *The Twelfth International Conference on Learning Representations*.
- [60] Haodong Duan, Junming Yang, Yuxuan Qiao, Xinyu Fang, Lin Chen, Yuan Liu, Xiaoyi Dong, Yuhang Zang, Pan Zhang, Jiaqi Wang, et al. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models. In *Proceedings of the 32nd ACM international conference on multimedia*, pages 11198–11201, 2024.
- [61] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Yuchen Duan, Hao Tian, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Yue Cao, Yangzhou Liu, Weiye Xu, Hao Li, Jiahao Wang, Han Lv, Dengnian Chen, Songze Li, Yanan He, Tan Jiang, Jiapeng Luo, Yi Wang, Cong He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Ying Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Lijun Wu, Kai Zhang, Hui Deng, Jiaye Ge, Kaiming Chen, Limin Wang, Mingsong Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhai Wang. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.
- [62] Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internvl3. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- [63] OpenAI. Introducing GPT-5.5. <https://openai.com/index/introducing-gpt-5-5/>, Apr 2026. OpenAI Blog.
- [64] Hugo Laurençon, Lucile Saulnier, Léo Tronchon, Stas Bekman, Amanpreet Singh, Anton Lozhkov, Thomas Wang, Siddharth Karamcheti, Alexander Rush, Douwe Kiela, et al. Obelics: An open web-scale filtered dataset of interleaved image-text documents. *Advances in Neural Information Processing Systems*, 36:71683–71702, 2023.

- [65] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? In *European conference on computer vision*, pages 216–233. Springer, 2024.
- [66] X.AI. Grok-1.5 vision preview. <https://x.ai/blog/grok-1.5v>, 2024.
- [67] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9556–9567, 2024.
- [68] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, et al. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15134–15186, 2025.
- [69] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023.
- [70] Yuliang Liu, Zhang Li, Mingxin Huang, Biao Yang, Wenwen Yu, Chunyuan Li, Xu-Cheng Yin, Cheng-Lin Liu, Lianwen Jin, and Xiang Bai. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12):220102, 2024.
- [71] bloc97. NTK-Aware Scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation., 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/141z7j5/ntkaware_scaled_rope_allows_llama_models_to_have/.
- [72] bloc97. Add NTK-Aware interpolation "by parts" correction, 2023. URL <https://github.com/jquesnelle/scaled-rope/pull/1>.
- [73] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- [74] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

Appendix

A Final Recipe and Implementation

A.1 Final LongPT Recipe

We summarize the final LongPT recipe used for the main results. The recipe is derived from the design choices studied in [Sections 4 and 5](#): we use long-document VQA as the primary source of synthesizing data, naturally sample training sequences over the target length range of 32K–128K, and use an 8 : 2 extraction-to-reasoning mixture. We list the full configuration of the final LongPT recipe in [Table 6](#).

Table 6 Final training recipe for MMProLong, which is selected from the ablation studies in [Sections 4 and 5](#).

Long-Context Continued Pre-Training (LongPT)		
Long Data	Data Synthesis	extract-single , extract-multi , and reasoning
	Distribution	<i>Pool-native</i> : natural sampling from the document pool over the target length range [32K, 128K]
	Mixture	8:2 extraction-to-reasoning ratio. The three task ratio is 40% extract-single , 40% extract-multi , 20% reasoning
Short Data	Default	None (pure long-context data)
	Alternative	60% long-context data, 40% LLaVA-OneVision data for better short-context preservation
Model	Initialization	Qwen2.5-VL-7B-Instruct (original mRoPE base freq. 1×10^6)
	mRoPE for 128K	4×10^6
	Maximum length	131,072 (128K tokens)
Optim.	Token Budget	5B tokens (2.9K H20 hours)
	Optimizer	AdamW (weight decay = 0.1, $\beta_1 = 0.9$, $\beta_2 = 0.95$)
	LR schedule	1×10^{-5} with 10% warmup and cosine decay to 1×10^{-6}
	Batch size	4M tokens (32 sequences)
	Framework	VeOmni with FlashAttention
	Parallelism	Sequence parallelism size 2 and FSDP size 4

A.2 Training Implementation Details

We conduct our LongPT experiments based on Qwen2.5-VL-7B [15], whose original context window is 32K, and extend it to 128K. For RoPE base frequency, we follow Dynamic-NTK [51] and set the mRoPE base frequency from the original value of 1×10^6 to 4×10^6 by default. We present the ablation study of the base frequency in [Appendix G.3](#). Then, we further apply the final training recipe to Qwen3-VL-8B [3] to test whether the recipe transfers across backbone models in [Appendix G.7](#). We train both models with VeOmni [58], a scalable framework for multimodal pre-training. The optimizer used is AdamW, using a peak learning rate of 1×10^{-5} , cosine decay to 1×10^{-6} , and 10% linear warmup. Each LongPT run is trained for a fixed budget of 5B tokens, with a maximum sequence length of 131,072 tokens (128K) and a global batch size of 4M tokens, which contains 32 sequences per update. Throughout the paper, we use binary prefixes: $K = 2^{10}$, $M = 2^{20}$, and $B = 2^{30}$.

To speed up, we use FlashAttention [59] for efficient attention computation in long-context data. In addition, Ulysses sequence parallelism size 2 and FSDP size 4 are used to fit the 128K training configuration on a single 8-GPU NVIDIA H20 node; in practice, we train with 8 H20 nodes (64 GPUs in total) to improve throughput.

B Full Evaluation Details

We describe the details of our evaluation. Here, for tasks from MMLongBench [5], including long-document VQA and MM-NIAH tasks, we follow their evaluation and use their released code v1.1². Meanwhile, we use the VLMEvalKit [60] to evaluate LVLMS on other tasks, including VTCBench, long video benchmarks, and short-context benchmarks.

B.1 Long-Document VQA

For the main long-document evaluation, we use the document category of MMLongBench [5], which contains MMLongBench-Doc [47], LongDocURL [48], and SlideVQA [55]. All examples in this category are instantiated at five standardized context lengths: 8K, 16K, 32K, 64K, and 128K tokens. Unless otherwise specified, we evaluate each model at 64K and 128K context lengths.

For grading, we follow the MMLongBench v1.1 evaluation protocol and report the official LLM-judged document QA score for each dataset. Specifically, this v1.1 protocol introduces LLM-based judging for the long-document VQA tasks, which handles different answer formats separately. For simple answer formats, such as string, integer, float answers, and “not answerable,” the judge assigns a binary score indicating whether the predicted answer matches the reference answer as shown in Table 7. For list-style answers, the evaluator first extracts the predicted list and then computes an F1 score based on the overlap between the predicted list and the reference list. We show this prompt in Table 8. For each context length, the AVG. column is the macro average over the three datasets, and the overall AVG. is the macro average over the 64K and 128K results.

B.2 Details of Long-Document VQA Evaluation with the Final Recipe

In evaluating **MMPROLONG** trained with the final recipe, we compare it with a wide range of open- and closed-source LVLMS. For open-source models, we evaluate Qwen2.5-VL (7B, 32B, 72B) [15], InternVL3 (8B, 14B, 38B) [61], InternVL3.5 (8B, 14B, 38B) [62], Gemma3 (4B, 12B, 27B) [52], and Gemma4 (E2B, E4B, 26B-A4B, 31B) [53]. For closed-source models, we include GPT-5.4 [13], GPT-5.5 [63], Gemini-2.5-Pro [23], Gemini-3.1-Flash [12], and Gemini-3.1-Pro [12] with a high reasoning budget.

B.3 Longer-Context Evaluation up to 512K

To test whether the 128K recipe extrapolates to longer contexts, we further evaluate the same three long-document VQA datasets at 256K and 512K. The examples at 256K and 512K context lengths are obtained following MMLongBench [5]: we alternately pad the left and right sides with randomly sampled negative documents until the required length is reached. No additional training or inference-time adaptation is applied to tested models when evaluating at these longer lengths. We use the same official LLM-judged score and macro-average computation as in the main long-document evaluation.

B.4 MM-NIAH: Evaluation on Webpage Haystacks

MM-NIAH [17] is a multimodal needle-in-a-haystack benchmark that builds on the webpages from OBELICS [64]. In particular, the benchmark contains three task families: retrieval, counting, and reasoning, each with two variants: text-needle and image-needle. In our evaluation, we use the standardized version provided by MMLongBench [5], where each example is instantiated at five context lengths: 8K, 16K, 32K, 64K, and 128K tokens. In our study, we report results at 64K and 128K context lengths as we extend the context window from 32K to 128K. For reporting, we first average the text and image variants within each task family. Then, the average (AVG.) score is the macro average over the three task-family scores. We use the metrics provided by MMLongBench [5] for each task variant, including exact match, soft accuracy, and multiple-choice accuracy, depending on the specific subtask definition.

²<https://github.com/EdinburghNLP/MMLongBench>

Table 7 Prompt for the binary answer judge in the long-document VQA category of MMLongBench. Gray highlighted spans denote placeholders filled at evaluation time.

Now your role is a grading teacher. Your task is to review and score student answers based on reference standard answers. You need to notice the following key points:

- First, extract the final answer from the student’s solution, then analyze and judge whether the answer is correct.
- Scoring should only refer to the final answer obtained by the student; there is no need to examine whether the intermediate problem-solving steps are correct.
- When analyzing and judging whether the answer is correct, you need to write down the scoring rationale, organize it into clear statements that follow the logical flow. The summary of the scoring rationale should be placed at the end, using the following format: "In summary, the student’s answer deserves x points" (where x represents the student’s specific score).
- Keep the whole process concise, within 150 words.
- Provide the score based on your analysis and display it in a code block in "JSON" format.
- An item is covered if it is strictly mentioned or unambiguously implied by a semantic equivalence. This includes numerical equivalence (e.g., 10% and 0.1), synonyms (e.g., UK and United Kingdom), and plural/singular forms (e.g., "apple" and "apples"). However, do not accept loosely related concepts.

Your output format is:

```
[Scoring Rationale]:
[Score]: x points
[JSON]:
{"answer_score": <integer_value>}
```

Below is the grading rubric:

```
[Scores]:
The scoring scale consists of 2 levels in total, from highest to lowest: 1 point, 0 points (the minimum is 0 points; if a situation arises where points need to be deducted beyond 0, simply assign 0 points).
[Tier Details]:
1 point: Assign 1 point if the student’s final answer matches the standard answer. If the question has multiple sub-questions, all sub-questions must be answered to assign 1 point.
0 points: Assign 0 points if the student’s final answer does not match the standard answer.
```

`<in-context exemplars_1>`

`<in-context exemplars_2>`

`<in-context exemplars_3>`

`<test_case_1>`

B.5 VTCBench: Evaluation on Long-Context Vision-Text Compression

VTCBench [18] is another multimodal long-context benchmark that evaluates whether a model can preserve visual-text information under compressed visual context. We follow the VTCBench-Wild setting reported in the original paper and report all three task scores: Retrieval, Reasoning, and Memory. We follow the same grading rules as the original paper. Retrieval and Reasoning are measured by the official `containsAll` accuracy, which checks whether all ground-truth answers are contained in the model prediction. Memory is measured by the official LLM accuracy, where `gpt-4o-mini` judges answer correctness. The AVG. score follows the Overall column of VTCBench-Wild and is the sample-count weighted average over the three splits, with 800 retrieval examples, 800 reasoning examples, and 600 memory examples.

B.6 Long-Video Understanding Evaluation

For long-video understanding, we report the performance on Video-MME [8], MLVU [19], and LongVideoBench [7]. We evaluate these benchmarks using the 1fps variants and cap the maximum number of sampled frames per video at 768, with the total number of video tokens not exceeding 24,576. This configuration is fully aligned with Qwen2.5-VL technical report [15]. For Video-MME and LongVideoBench, we report overall multiple-choice accuracy. For MLVU, we evaluate the multiple-choice subset and report its average accuracy.

Table 8 Prompt for the list-style answer judge in the long-document VQA category of MMLongBench.

Now your role is a grading teacher. Your task is to review and score student answers for LIST-style questions, where the standard answer is a list of required items.

- First, extract the specific list of items from the <Student Answer>. Ignore conversational filler (e.g., "The answer is...").
- Then, compare the [Extracted List] against the <Standard Answer> (Ground Truth).

Here are some extra key points:

- The standard answer is a JSON-like list of items with each item as one required element. Determine whether each item is covered by the student's answer list.
- An item is covered if it is strictly mentioned or unambiguously implied by a semantic equivalence. This includes numerical equivalence (e.g., 10% and 0.1), synonyms (e.g., UK and United Kingdom), and plural/singular forms (e.g., "apple" and "apples"). However, do not accept loosely related concepts.
- You need to write down the extraction and comparing rationale, organize it into clear statements that follow the logical flow. The summary of the rationale should be placed at the end, using the following format: "In summary, the student's answer list has X items, covering Y items from the reference list."
- Keep the whole process concise, within 200 words.
- Provide the student's answer item count and covered item count in a code block in "JSON" format.

Your output format is:

[Rationale]:

[JSON]:

```
{
  "student_answer_count": <integer_value>,
  "covered_count": <integer_value>
}
```

<in-context exemplars_1>

<in-context exemplars_2>

<in-context exemplars_3>

<test_case_1>

B.7 Short-Context Evaluation

To monitor whether **MMPROLONG** suffers degradation in general VLM ability, we evaluate short-context performance across three capabilities and six benchmarks. These include general VQA (MMBench-V1.1 [65] and RealWorldQA [66]), multimodal reasoning (MMMU [67], MMMU-Pro [68], and MathVista [69]), and text recognition (OCRBench [70]).

For MMMU, we use `MMMU_DEV_VAL` in `VLMEvalKit`, which combines the official MMMU development and validation splits. For MMMU-Pro, we use `MMMU_Pro_10c`, the standard 10-choice variant. For MathVista, we use the testmini subset. We report the official score of each benchmark and compute the short-context AVG. as the macro average over the six benchmark scores.

C Preliminary Details

C.1 Document Pool Statistics

We show the corpus scale, average page length, and language distribution of our document pool in Table 9. Meanwhile, the distribution of the document domain is shown in Figure 6.

C.2 OCR Expert

We use an OCR expert model fine-tuned from Seed 2.0 [4] to preprocess the rendered PDF pages. For each page image, the OCR expert parses layout-aware text blocks and assigns structural labels such as title, section heading, paragraph, table, figure caption, header, and footer.

These parsed blocks serve two purposes in our pipeline: (i) the title and section labels provide lightweight

document-structure signals for sampling semantically coherent page spans when generating long-document VQA data; (ii) the recognized text blocks provide the target text for constructing OCR transcription baselines.

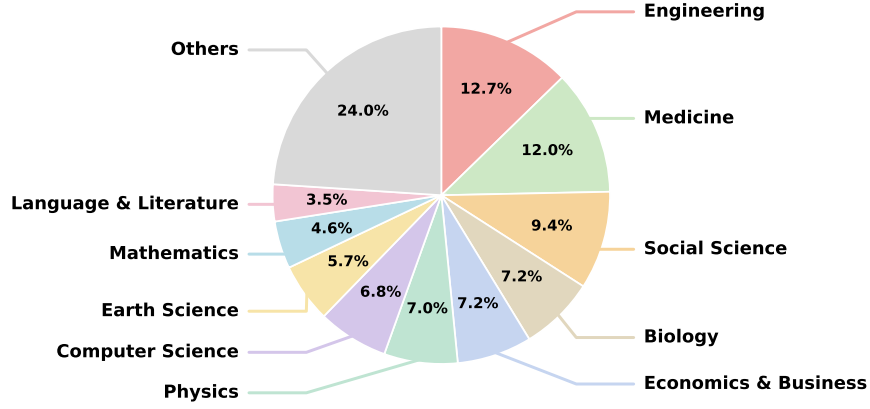


Figure 6 The document domain distribution of the document pool used for data synthesis.

Table 9 Statistics of the document pool used for data synthesis.

Category	Statistic	Value
Corpus Scale	Number of documents	1,537,504
	Page-count range	[20, 200]
	Average Pages	23.80
	Total pages	36,592,809
Lang.	English documents	1,479,370 (96.22%)
	Chinese documents	55,202 (3.59%)
	Other languages	2,932 (0.19%)

D Long-Document VQA Training Data Details

D.1 Data Statistics for Pool-Native Distribution (Default)

We summarize the statistics of the long-document VQA training data in Table 10a, with the corresponding token-length distribution shown in Figure 7a. This data corresponds to the *pool-native* length distribution used by our final LongPT recipe, covering samples from 32 to 50 rendered PDF pages and approximately 32K–128K multimodal tokens.

D.2 Data Statistics for Long-Biased Distribution

We summarize the statistics of the *long-biased* long-document VQA training data in Table 10b. The corresponding token-length distribution is shown in Figure 7b.

Compared with the default *pool-native* distribution, this data is sampled from longer documents with 50–100 rendered PDF pages and concentrates more training mass near the upper end of the target context window. As shown in Table 11, 83.9% of the samples in the *long-biased* distribution contain at least 100K tokens across

Table 10 Statistics of the long-context training data used in our data-design study.

(a) Long-document VQA data in the <i>pool-native</i> Distribution (default).						
Task	# Samples	# Pages	# Tokens (K)	Total Tokens (B)	Page Range	Token Range (K)
extract-single	59,055	38.4	85.3	5.04	[32, 50]	[32.8, 126.8]
extract-multi	59,316	38.4	85.3	5.06	[32, 50]	[32.8, 126.8]
reasoning	59,403	38.4	85.3	5.06	[32, 50]	[32.8, 126.7]
Total	177,774	38.4	85.3	15.16	[32, 50]	[32.8, 126.8]
(b) Long-document VQA data in the <i>long-biased</i> distribution.						
Task	# Samples	# Pages	# Tokens (K)	Total Tokens (B)	Page Range	Token Range (K)
extract-single	59,736	66.8	114.3	6.83	[50, 100]	[32, 125.3]
extract-multi	59,624	66.8	114.3	6.81	[50, 100]	[32, 124.0]
reasoning	59,770	66.8	114.3	6.83	[50, 100]	[32, 126.7]
Total	179,130	66.8	114.3	20.47	[50, 100]	[9.2, 126.7]
(c) OCR transcription data.						
Task	# Samples	# Pages	# Tokens (K)	Total Tokens (B)	Page Range	Token Range (K)
OCR-full	97,336	39.6	96.4	9.38	[32, 50]	[32.8, 122.9]
OCR-needle	140,655	42.9	85.4	12.01	[32, 50]	[32.8, 122.9]
Total	237,991	41.6	89.9	21.39	[32, 50]	[32.8, 122.9]

Table 11 Percentage of training samples with token length at least 100K for both *pool-native* and *long-biased* distribution.

Distribution	extract-single	extract-multi	reasoning
pool-native	23.6%	23.6%	23.5%
long-biased	83.9%	83.9%	83.9%

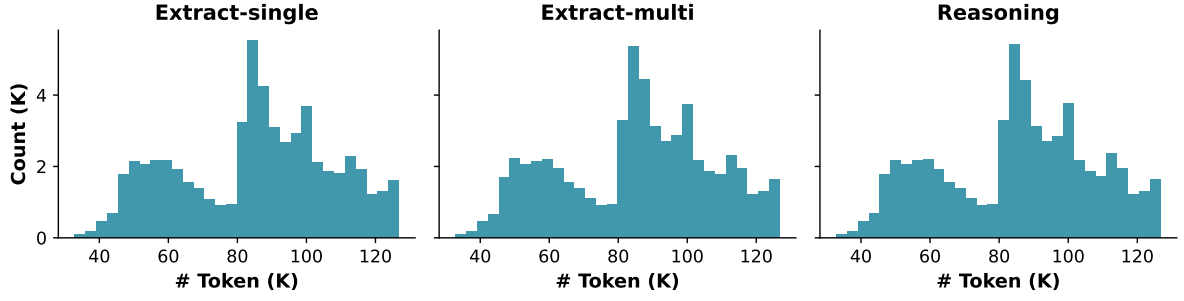
all three VQA tasks. In contrast, only about 23.5%–23.6% of the samples in the *pool-native* distribution exceed this threshold.

D.3 Prompt Template for QA Pair Generation

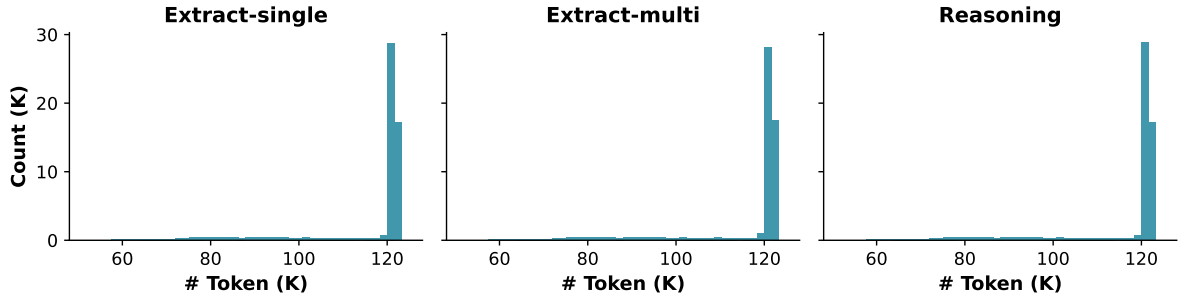
For each source document, we first use OCR block labels to identify title and section boundaries, then sample a semantically coherent span of 8–15 consecutive pages. The sampled page images are sent to the teacher LVLM together with a prompt that asks it to synthesize one document-grounded QA pair.

By default, we include two in-context exemplars sampled from a small manually curated exemplar pool. We instantiate the same core prompt with different task descriptions and page constraints for three data types: single-page extraction, multi-page extraction, and reasoning.

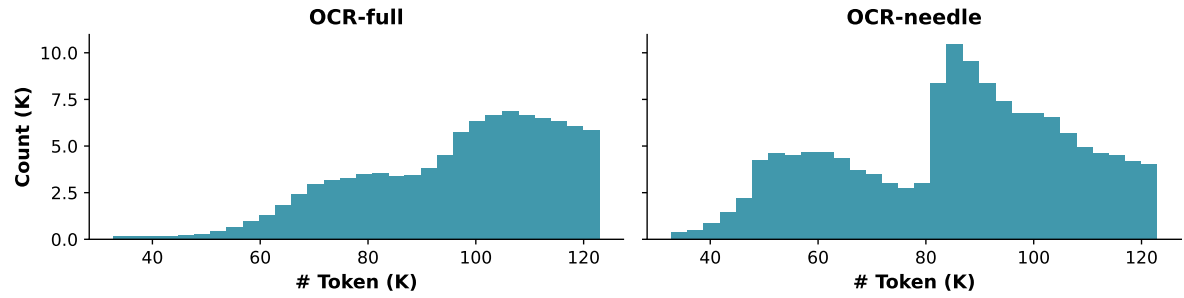
More concretely, Table 12 shows the prompt template used in our data sourcing process, and Table 13 summarizes the task description and extra restriction inserted for each task type.



(a) Long-document VQA data in the *pool-native* Distribution (default).



(b) Long-document VQA data in the *long-biased* distribution.



(c) OCR transcription data.

Figure 7 Token-length distributions of the long-context training data used in our data-design study.

Table 12 The prompt template used for long-document QA pair generation. Gray highlighted spans denote placeholders filled dynamically for each sampled document span.

[System]
You are an expert in synthesizing document question-answering dialogue. In this task, I will provide you the images of one or more sections from a document and you need to generate a question-answering pair based on the given sections. You need to notice the following key points:

[Task Definition & Requirements]
The current document QA task is `task description`.

[General Restrictions]
- Questions must be confidently answerable and document-dependent, relying solely on the provided visual content without external knowledge.
- If the question asks about missing information, the absence of such content must be definitively verifiable from the provided image.
- Scope Constraint: Frame questions specifically within the scope of the provided section/pages, strictly avoiding broad references like "across the document" or "the whole file."
- Page Indexing Rule: When writing the evidence description or the "evidence_pages" field in JSON, strictly use the index provided in the prompt (e.g., use '10' for "Page 10"), regardless of the page number inside images.
`extra_restriction`

[Response Format]
Please generate the response in two parts:

Part 1: Evidence Description
Provide a detailed description first, citing specific text or visual elements (using the four evidence types defined below) to support your reasoning. You may use multiple paragraphs if necessary.

When describing evidence, explicitly categorize the text and visual elements into one of the following four types:

- Text: pure texts, such as paragraphs.
- Layout: text elements with special layout meaning (generalized text), such as titles, headers, footers, table names, and figure names.
- Figure: including charts and general images.
- Table: structured data in rows and columns.

Part 2: JSON Output
Provide the final question, answer, and metadata in a strict JSON format.

Your answer must strictly fall into one of the following four categories. You must also indicate the type in the JSON output:

1. String: General text, names, short sentences, or short phrases found in the document.
- Example: "John Doe", "Financial Report", "The project was completed on time."
2. Integer: Whole numbers representing counts, years, page numbers, etc.
- Example: "42", "2023", "100".
3. Float: Numbers with decimal points, including currency, percentages, or scientific measurements.
- Example: "12.5", "\$45.20", "98.5%".
4. List: A collection of multiple items, names, or values.
- Example: ["Apple", "Banana", "Orange"], ["Item A", "Item B"].

Your output format is:

[Evidence Description]:
[JSON]:
{"question": <question>, "answer": <answer>, "answer_format": <answer_format>, "evidence_pages": [<page_index>], "evidence_sources": [<evidence_sources>]}
<>

`<in-context exemplars_1>`
`<in-context exemplars_2>`

[Current Case]
The following images represent pages `start_page` to `end_page` of the document. Generate the question-answering pair strictly based on the visual content below, ensuring the question scope is limited to phrases like "the Introduction section", "Pages 20, 21, and 25", or "Page 10."
`current_case_image_sequence`

Table 13 The task descriptions and extra restriction for the three long-document VQA tasks. We insert these parts into the prompt template to synthesize data.

Task	Task Description	Extra Restriction
extract-single	information extraction. You need to generate questions that focus on extracting specific, explicit information directly from the provided document images. Your objective is to create questions that ask for precise facts, entities (such as names, dates, locations, or authors), numerical values, lists of items, or specific steps in a procedure. The answers must be visually present and directly retrievable from the text, tables, or layout elements.	Single-Page Focus: Although multiple pages are provided for context, you must generate a question that is **strictly self-contained within a single page** . Select one specific page from the provided sequence and generate a question based solely on its content. The answer must be fully derivable from that single page without requiring cross-referencing or synthesizing information from other pages.
extract-multi	information extraction. You need to generate questions that focus on extracting specific, explicit information directly from the provided document images. Your objective is to create questions that ask for precise facts, entities (such as names, dates, locations, or authors), numerical values, lists of items, or specific steps in a procedure. The answers must be visually present and directly retrievable from the text, tables, or layout elements.	Multi-Page Priority: When multiple pages are provided, **prioritize** generating questions that require synthesizing information across different pages (≥ 2 pages). For example, link a table header from one page to a row on the next, or aggregate data points scattered across multiple pages. You can also come up with other formats of multi-page questions. **Only fall back to single-page questions if no meaningful cross-page connections exist.**
reasoning	quantitative reasoning. You need to generate questions that require performing calculation, comparison, or counting. Your objective is to create questions involving one of the following: 1. Calculate: Performing arithmetic operations on data found in the document. 2. Compare: Comparing values to identify trends, maximums, minimums, or relative sizes. 3. Count: Counting the frequency of specific items, keywords, or visual elements that satisfy a condition. The answer must require the aforementioned processing step rather than being directly visible as a single contiguous string.	- Non-Trivial Synthesis: The question must require **aggregating multiple data points** . For example, ask “What is the difference in revenue between Q1 and Q2?” or “Calculate the total sum of items listed in Table 3.” The key is that the user must find at least two pieces of information and combine them to get the answer.

D.4 Human Verification of QA Pairs

We randomly sample 100 generated QA pairs from the three task types and manually verify their answer correctness and evidence consistency. Among these inspected examples, 97 are fully correct: two contain incorrect answers, and one has an inaccurate evidence annotation. This suggests that the synthesis pipeline produces high-quality document-grounded supervision overall, while also indicating that the generated data may still contain a small amount of noise.

E OCR Transcription Details

In addition to VQA-based synthesis, we construct OCR transcription data as a contrasting family of long-context supervision. Each training example places rendered document pages as visual input and asks the model to generate the corresponding OCR text parsed from those pages. We consider two variants. Full-document OCR requires transcribing text elements from all pages of the document, creating dense image-text alignment across the full context. Needle-page OCR keeps the long visual context but only asks the model to transcribe text from a small subset of selected pages, making the task closer to retrieval over many distractor pages.

E.1 Data Statistics

We summarize the statistics of the OCR transcription data in Table 10c. The corresponding token-length distribution is shown in Figure 7c.

F Short-Context Data Details

We use short-context data in two sets of experiments. First, in the comparison between long-document VQA and OCR transcription data (Section 4.4), we additionally apply supervised fine-tuning (SFT) with short-context instruction data after training the model on OCR transcription data. Second, in the short-data mixture experiment (Section 5.3), we mix short-context data with long-context data to study how much short-context supervision should be included in LongPT.

For both settings, we use publicly released short-context instruction data from LLaVA-OneVision [54]. We use the full short-context instruction data in the short-data mixture experiment, while only the SFT portion is used in the additional SFT stage after OCR transcription training. We download the single-image and multi-image subsets from the LLaVA-OneVision Hugging Face repositories³⁴.

G Complementary Experimental Results

G.1 Sequence-Length Distribution of the Training Data

We provide the full dataset-level results for the sequence-length distribution ablation in Table 14.

Table 14 We compare the *pool-native* and *long-biased* distributions under each of the three VQA-based training tasks. We abbreviate these three datasets as MMLB-D, LD-URL, and SLIDE, respectively.

Training data	Length	64K MMLongBench				128K MMLongBench				AVG.
		MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
extract-single	pool-native	33.85	59.73	77.00	56.86	30.89	55.69	77.00	54.53	55.69 +1.3
	long-biased	31.47	55.88	75.00	54.12	29.48	59.40	75.00	54.63	54.37
extract-multi	pool-native	32.75	64.32	77.00	58.02	31.50	54.82	81.00	55.77	56.90 +0.1
	long-biased	36.29	56.75	80.00	57.68	30.72	55.79	81.00	55.84	56.76
reasoning	pool-native	32.67	60.34	79.00	57.33	29.23	61.61	76.00	55.62	56.47 +1.7
	long-biased	32.00	55.79	79.00	55.60	28.74	58.22	75.00	53.98	54.79

G.2 Short-Data Mixing for Long-Document VQA

We provide the full per-dataset long-document VQA results for the short-data mixing ablation in Table 15, complementing the AVG-only summary plotted in Figure 3.

³<https://huggingface.co/datasets/lmms-lab/LLaVA-OneVision-Data>

⁴<https://huggingface.co/datasets/lmms-lab/M4-Instruct-Data>

Table 15 Short-context data test. We mix in 0% to 80% short-context data during LongPT under the fixed 5B-token budget. The long-context data uses the 8:2 extraction-to-reasoning long-context mixture. **Short Data** means the proportion of short-context data. 0% means only using long-context data.

Short Data	64K MMLongBench				128K MMLongBench				AVG.
	MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
0%	36.00	62.69	80.00	59.56	34.19	56.33	77.00	55.84	57.70
20%	32.33	57.61	78.00	55.98	31.49	56.00	78.00	55.16	55.57
40%	34.90	56.65	80.00	57.19	30.19	61.30	79.00	56.83	57.01
60%	33.62	57.53	81.00	57.38	31.93	58.61	79.00	56.52	56.95
80%	33.90	57.60	80.00	57.17	31.93	55.17	81.00	56.03	56.60

G.3 mRoPE Base Frequency

Existing studies [25, 34, 71, 72] have shown that increasing the RoPE [73] frequency base during long-context continued pre-training or inference can improve long-context performance. Dynamic-NTK [51] suggests scaling the frequency base by $t^{\frac{d}{d-2}}$, where t denotes the context-window expansion factor and d is the attention head dimension.

However, LVLMs often use more structured positional encodings than plain 1-D RoPE. In our experiments, Qwen2.5-VL-7B adopts mRoPE [74], which decomposes rotary embeddings into temporal, height, and width components. As a result, visual position indices grow more slowly than a flattened 1-D sequence. It is therefore unclear whether the RoPE-scaling heuristic from Dynamic-NTK, originally developed for LLMs, directly applies to LVLMs with mRoPE.

We conduct an ablation study over the mRoPE frequency base for LVLm long-context training. By default, we follow Dynamic-NTK and scale the frequency base of the trained model from 1×10^6 to 4×10^6 when extending the context window from 32K to 128K tokens. We also evaluate two alternative bases, 2×10^6 and 8×10^6 , to examine whether decreasing or increasing the default 4×10^6 base improves LongPT performance.

The evaluation results on long-document VQA tasks are shown in Table 16. Overall, moderately increasing the mRoPE base improves or maintains long-context performance compared with using a smaller base. Across the evaluated tasks, 2×10^6 and the Dynamic-NTK-scaled base 4×10^6 achieve comparable overall performance, with 4×10^6 slightly outperforming 2×10^6 on **extract-multi** and **reasoning**. Further increasing the base to 8×10^6 improves some individual metrics, but does not yield consistent gains across tasks and can degrade performance on **extract-multi** and **reasoning**. These results suggest that moderate mRoPE-base scaling is sufficient for extending LVLMs to longer contexts, while overly aggressive scaling is unnecessary. Based on this observation and to maintain consistency with the Dynamic-NTK heuristic, we set 4×10^6 as the mRoPE base in our main experiments.

Table 16 Ablation on the mRoPE frequency base for 128K LongPT. We grid search the mRoPE base in $\{2 \times 10^6, 4 \times 10^6, 8 \times 10^6\}$, starting from the original base of 1×10^6 used for the 32K context. The Dynamic-NTK heuristic [51] suggests 4×10^6 as the default base when extending the context window from 32K to 128K. Overall, 2×10^6 and 4×10^6 achieve comparable performance, while further increasing the base to 8×10^6 does not yield consistent gains across tasks.

Training data	Freq.	64K MMLongBench				128K MMLongBench				AVG.
		MMLB-D	LD-URL	SLIDE	AVG.	MMLB-D	LD-URL	SLIDE	AVG.	
extract-single	2×10^6	35.52	57.07	78.00	56.86	28.87	58.65	78.00	55.17	56.02
	4×10^6	33.85	59.73	77.00	56.86	30.89	55.69	77.00	54.53	55.69
	8×10^6	33.63	52.73	79.00	55.12	30.83	58.10	79.00	55.98	55.55
extract-multi	2×10^6	31.86	63.01	76.00	56.95	31.60	52.31	79.00	54.30	55.63
	4×10^6	32.75	64.32	77.00	58.02	31.50	54.82	81.00	55.77	56.90
	8×10^6	33.36	58.91	79.00	57.09	29.25	54.00	81.00	54.75	55.92
reasoning	2×10^6	33.04	57.20	77.00	55.75	33.44	58.66	77.00	56.36	56.05
	4×10^6	32.67	60.34	79.00	57.33	29.23	61.61	76.00	55.62	56.47
	8×10^6	34.33	57.86	73.00	55.07	34.26	54.61	72.00	53.62	54.34

G.4 MM-NIAH Generalization

We provide the full 64K and 128K MM-NIAH results in Table 17, complementing the averaged main-text summary in Figure 4.

Table 17 MM-NIAH generalization. Ret., Count, and Reas. average text and image needles.

(a) 64K MM-NIAH					(b) 128K MM-NIAH				
Model	Ret.	Count	Reas.	AVG.	Model	Ret.	Count	Reas.	AVG.
MMPROLONG	74.83	27.67	67.33	56.61	MMPROLONG	57.83	8.67	60.33	42.28
Qwen2.5-VL-7B	50.00	6.00	27.50	27.83	Qwen2.5-VL-7B	11.33	16.33	8.83	12.17
InternVL3-8B	67.67	14.33	59.33	47.11	InternVL3-8B	52.33	7.33	50.00	36.56
InternVL3.5-8B	56.12	4.45	57.23	39.26	InternVL3.5-8B	3.45	0.00	2.27	1.91
Gemma3-4B	43.25	4.33	31.50	26.36	Gemma3-4B	29.83	1.83	28.75	20.14
Gemma3-12B	65.58	27.00	42.25	44.94	Gemma3-12B	48.83	19.67	33.75	34.08

G.5 VTCBench Generalization

We provide the full VTCBench-Wild results in Table 18. We report retrieval, reasoning, memory, and the sample-count weighted overall score following VTCBench-Wild.

Table 18 VTCBench generalization. We report retrieval, reasoning, memory, and the sample-count weighted overall score following VTCBench-Wild.

Model	Ret.	Reas.	Mem.	AVG.
MMPROLONG	91.75	22.88	40.50	52.73
Qwen2.5-VL-7B	91.63	15.63	33.83	48.23
Qwen3-VL-8B	89.00	11.50	33.67	45.73
InternVL3.5-8B	51.38	7.63	17.33	26.18
InternVL3.5-38B	45.81	8.75	22.33	25.93
Gemma3-27B	49.38	3.75	17.33	24.05

G.6 Long-Video Generalization

We provide the numeric long-video generalization results in Table 19, complementing the main-text summary in Figure 5.

Table 21 MM-NIAH transfer across backbone models. We report Retrieval, Counting, Reasoning, and average scores at 64K and 128K context lengths. Our recipe achieves better overall MM-NIAH score on Qwen3-VL-8B.

Model	64K MM-NIAH				128K MM-NIAH				AVG.
	Ret.	Count	Reas.	AVG.	Ret.	Count	Reas.	AVG.	
Qwen3-VL-8B	77.83	25.33	62.33	55.17	61.67	17.00	56.00	44.89	50.03
+ MMPProLONG Recipe	83.00	45.67	70.83	66.50	74.17	35.00	61.83	57.00	61.75 +11.7

Table 19 Long-video generalization. We report aggregate scores on Video-MME, MLVU, and LongVideoBench.

Model	Video-MME	MLVU	LongVideoBench
Qwen2.5-VL-7B	65.1	70.2	60.43
MMPProLONG	67.78	73.55	62.08

Table 20 Long-document VQA transfer across backbone models. We apply LongPT variants to Qwen3-VL-8B and report results on LongDocURL and SlideVQA. Since Qwen3-VL is already a 256K-context model trained with large-scale long-context pre-training, SFT, and RL, we focus on these two held-out document benchmarks.

Model	64K MMLongBench			128K MMLongBench			AVG.
	LD-URL	SLIDE	AVG.	LD-URL	SLIDE	AVG.	
Qwen3-VL-8B	56.36	74.00	65.18	60.11	72.00	66.05	65.62
+ MMPProLONG Recipe	62.22	73.00	67.61	62.81	72.00	67.41	67.51 +1.9

G.7 Generalization across Backbones

We further apply our LongPT recipe to Qwen3-VL-8B [3] to test whether the training recipe transfers beyond Qwen2.5-VL. However, the Qwen3-VL series already includes 256K-context models trained with 100B tokens in long-context continued pre-training, and Qwen3-VL has undergone additional SFT and RL optimization for long-document tasks [3]. Therefore, this experiment is not intended as a strict study on context window extension. Instead, we use it as a diagnostic to examine whether the observed behavior transfers across backbones. We report the performance on long-document VQA tasks and MM-NIAH in Tables 20 and 21, respectively.

The results show that the final **MMPProLONG** recipe remains effective even on this stronger long-context backbone. On long-document VQA, **MMPProLONG** improves the average score from 65.62 to 67.51. The gains are more pronounced on MM-NIAH, where the average score increases from 50.03 to 61.75, with consistent improvements across retrieval, counting, and reasoning at both context lengths. Together with the diagnostic nature of this experiment, these results suggest that the proposed training recipe is not specific to Qwen2.5-VL-7B, but may also improve the long-context behavior of a newer backbone that already incorporates native long-context training.

H Limitations

For the training scale, our systematic study is primarily conducted on 7B/8B-scale LVLMS. This choice enables controlled comparisons across data recipes, context lengths, and training budgets, but it also leaves open how the observed trends scale to substantially larger models. Extending the same study to 30B or 70B-scale LVLMS, or to even longer context windows, like 512K or 1M, would require significantly higher computational cost, since long-context continued pre-training is expensive both in model size and sequence length. Although our transfer experiment on Qwen3-VL-8B provides initial evidence that the proposed recipe is not tied to a single backbone, a more comprehensive scaling study across larger model families remains an important direction for future work.

For evaluation, our long-document VQA experiments rely on model-based judging to assess answer correctness. Compared with lexical-overlap metrics, such judges can better handle semantically equivalent answers and free-form generation, but they also introduce additional API cost. This cost becomes substantial when evaluating many checkpoints, context lengths, and model variants, which limits the frequency and scale of evaluation during long-context training. Developing more efficient, reliable, and low-cost evaluation protocols for multimodal long-context models is, therefore, an important direction for future research.

I Broader Impact

Reliable long-context capability is important for deploying LVLMs in real-world scenarios that require understanding and reasoning over large multimodal inputs, such as long documents, webpages, videos, and agentic workflows. This work studies how to build such capability through long-context continued pre-training, with a particular focus on constructing effective multimodal long-context data under a modest training budget. Our findings suggest that carefully designed long-document VQA data can provide useful supervision for evidence retrieval and reasoning over long visual-textual contexts, and can transfer beyond documents to broader multimodal long-context tasks. We hope these results can help the community develop more data-efficient training recipes for LVLMs, reducing the need for excessive token budgets while improving models' ability to use long multimodal context.